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ISSN: 0016-786X (Print) (Online) Journal homepage: http://www.tandfonline.com/loi/sgff19

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To cite this article: Sten-ÅKe Elming (1985) A palaeomagnetic study of Svecokarelian basic rocks from northern Sweden, Geologiska Föreningen i Stockholm Förhandlingar, 107:1, 17-35, DOI: 10.1080/11035898509452608

To link to this article: http://dx.doi.org/10.1080/11035898509452608

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A palaeomagnetic study of Svecokarelian basic rocks from northern Sweden

STEN-ÅKE ELMING

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Elming, S.-Å., 1985 11 30: A palaeomagnetic study of Svecokarelian basic rocks from northern Sweden. *Geologiska Föreningens i Stockholm Förhandlingar*, Vol. 107, Pt. 1, pp. 17–35. Stockholm. ISSN 0016-786X.

This palaeomagnetic investigation comprises basic rocks from six localities from the Svecokarelian zone in northern Sweden. Most of the pole positions in this study and other reported poles of Svecokarelian and post-Svecokarelian rocks fall within an approximately 12 degrees wide band running from east to west representing ages of magnetizations from 1880–1700 to ~1530 Ma. Thermal demagnetizations of specimens of the probably oldest massifs indicate a possible backward continuation of the polar wandering path. Mineralogical studies of thin sections of the rocks show ore symplectites and myrmekitic textures indicating a slow rate of cooling at least at the end of the rock formation. Signs of metamorphism are demonstrated by the existence of secondary minerals, including magnetite, not related to late magmatic alterations. The distribution of site means as well as the change of directions of the remanence vectors during thermal demagnetization can be explained by a slow rate of cooling and where signs of metamorphism exist by partial remagnetization of the rock. This study has, apart from the palaeomagnetic results, demonstrated the difficulty of correlating radiometric ages with ages of magnetization. □ *Svecokarelian, palaeomagnetism, AF-demagnetization, thermal-demagnetization, physical properties, mineralogy, basic rocks, Haparanda Suite, Nottträsk, Sangis, Tärendö, Svappavaara, Vittangi, Dundret, north Sweden, N6550, N6740, E2035, E2330.*

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Palaeomagnetic studies of Precambrian rocks are often very complicated because of the great likelihood that these rocks have been exposed to intense repeated metamorphism and thereby totally or partly remagnetized. Therefore the magnetization is often not easily relatable to the age determination. The emphasis of this study is on getting a better idea of these problems and if possible relate the magnetizations of some Swedish Precambrian rocks to specific ages.

General geology

The rocks in the northern part of Sweden are predominantly Svecokarelian granites. Pre-Svecokarelian rocks, mainly granitic gneisses, exist as isolated cores and constitute the basement of the supracrustal rocks. The minimum age of these granitic gneisses is 2800 Ma (Welin et al. 1971; Skiöld 1979) and they belong to the basement complex exposed in eastern Finland and Karelian part of the Soviet Union (Lundqvist 1979).

The formation of the supracrustal rocks is characterized by 'incipient rifting and by the intrusion and extrusion of basic magmas' (Witschard 1979). During this episode quartzitic sediments (the Tjärro formation) were deposited and the Kiruna and Vittangi greenstones formed (more than 2000 Ma ago). The supracrustals were thereafter intruded by primorogenic plutonic rocks represented by the Haparanda suite, gabbros, granodiorites and granites (~1840 Ma, Welin et al. 1970).

New radiometric age determinations of the volcanic rocks from the Kiruna area indicate these rocks (the Kiruna Porphyry) to be of similar age as the Haparanda suite. The predominance of the porphyry group in the west and the Haparanda suite in the east is thought to reflect different tectonic regimes existing in these regions (Witschard 1979). During the following culminating metamorphic phase the supracrustal and primorogenic rocks were deformed to various degrees and reworked. After a process of migmatization the serorogenic granites represented by, e. g. the Older Lina Granite (1780 Ma, Welin et al. 1971) intruded. Post-orogenic intrusions, here represented by the Younger

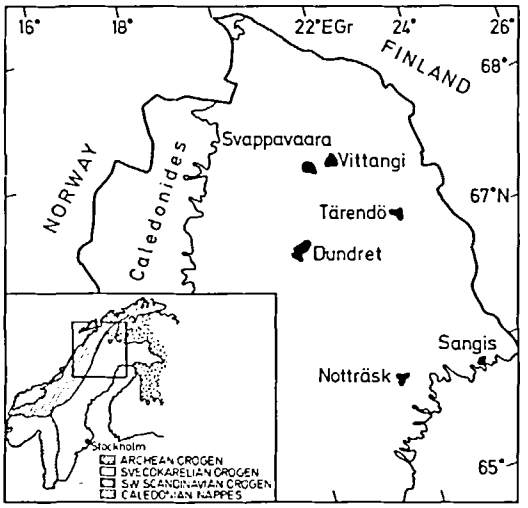


Fig. 1. A general geological subdivision of the Fennoscandian shield and the area of the present study. The sampled formations are marked in black.

Lina Granite (1530 Ma, Welin et al. 1971), are the youngest rocks in this part of the Baltic shield.

Field sampling and laboratory analyses

Some 230 samples from basic rocks probably belonging to the primorogenic Haparanda suite were taken for this study. Six different formations, located between 65.8 and 67.7 degrees northern latitude, were sampled, comprising the basic rocks at Notträsk, Sangis, Tärendö, Svappavaara, Vittangi and Dundret (Fig. 1). The samples were obtained with a portable rock drill and oriented with sun and magnetic compasses. About 5 samples were taken from each site covering at least 20–30 m^2 of outcrop.

The specimens were measured in a nonmagnetic laboratory (Parasnis & Elming 1982) with an astatic magnetometer and progressively demagnetized in alternating fields (AF) up to 55 mT. From each rock formation thermal demagnetizations were carried out on a few specimens. These specimens, demagnetized in a 55 mT AF strength, were chosen among those with stable directions representative of the formation. The thermal analyses are another test of the stability of the remanence and a further attempt to separate eventually different components of magnetization. The specimens were all treated in temperatures up to 600° in steps of 10° to 20°C when approaching the Curie temperature.

To reduce the possibility of oxidation of the specimens these thermal demagnetizations were carried out in argon gas. Temperature calibration experiments in the TSD-1 furnace used indicate the temperatures estimated to be accurate to ± 10°C.

Statistical analyses

Fisherian statistics are used to estimate the mean directions of the sites and of the formation. To avoid situations involving subjective judgement, determinations of mean directions have been carried out during the whole process of demagnetization. Together with the demagnetization graphs and stability of directions the precision parameter shows where to pick the most accurate formations' mean.

Geology, age and some physical properties of the sampled formations

An empirical relation between density and susceptibility of basic rocks of various compositions from the north of Sweden was established by Henkel (1976). If the susceptibility is attributed entirely to a content of magnetite, it is possible to estimate an equivalent 'silicate density' of the rock by reducing the effect of this content on the total density. By plotting the densities and susceptibilities these relations make a rough geological classification possible. The Q-value (Königsberger ratio)-susceptibility diagrams give an indication of the homogeneity in magnetic properties and together with the distribution of density possibly reflect differences in the process of rock genesis.

The *Notträsk gabbro,* probably belonging to the Haparanda suite (Ödman 1957), is a differentiated norite–olivine gabbro intruded in a sedimentary gneiss (Widenfalk et al. in press). The differentiation of the rocks is well recognized in the density variations as well as in the magnetic properties. An increase of iron and titanium outward from the centre of the formation has resulted in a strongly magnetic ringstructure. In the southern marginal areas the rock is partly metamorphosed and biotite and pyroxene are altered to mainly chlorite, ilmenite and some magnetite. The 20 km^2 large formation has been sampled at nine sites, four of which are situated in the

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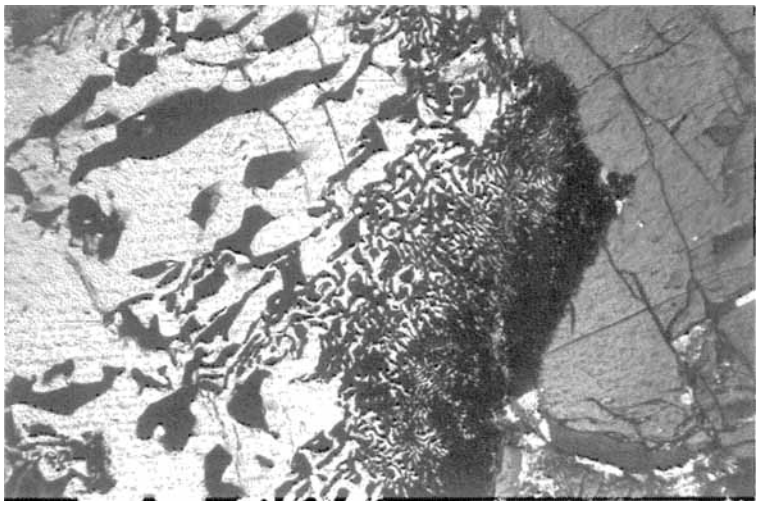


Fig. 2. Example of ore symplectite of magnetite from the Tärendö gabbro.

strongly magnetic zone. The Q-values are high, generally varying between 1.5 and 8.0.

The massif of *Sangis*, sampled at 12 sites, is the smallest of the rock formations in this study covering an area of 11 km^2. The rock is medium-grained, with silicate densities corresponding to compositional variations from diorite to ultrabasic rocks. Alteration has taken place and secondary magnetite as well as haematite, a consequence of martitization, exist. The rock is radiometrically dated to 1840 ± 390 Ma (Rb–Sr, Welin et al. 1970) and defined to be a part of the Haparanda suite. Serorogenic granites located not so far from the formation are dated to 1772 Ma (Rb–Sr, Skiöld 1977).

The irregulary shaped *Tärendö* gabbro, 45 km^2 in area, is a medium to coarse-grained massive gabbro. Chemical analyses show a normal gabbroic composition (Padget 1970), while silicate densities correspond to a rock of dioritic to mafic composition. A slow cooling rate for the rock is indicated by the presence of frequent ore-symplectites (Fig. 2). The gabbro is surrounded by a younger granite and has been affected by it to a varying degree. However, the microscope studies of specimens from four different sites show a fairly low degree of alteration and secondary magnetite occurs only in very small amounts. A minimum age of the gabbro is indicated by the Rb–Sr age of surrounding Lina (1780 ± 49 Ma, Welin et al. 1971) and Vettasjärvi (1757 ± 43 Ma, Skiöld 1982) granites.

The susceptibility and Q-values are generally quite high varying between 0.04 to 0.13 and 0.8 to 8.0 respectively.

The *Svappavaara* massif, 36 km^2 in area, is an irregularly shaped medium-grained rock with a dioritic to gabbroic composition. The quite high amount of magnetite occurs as primary as well as secondary mineral, but the secondary phase is probably a result of late magmatic alteration. Some myrmekite textures are also observed. The rock is quite inhomogeneous with pockets of granites, which probably are differentiates of the gabbro.

Three rock complexes surrounding and partly separating the Svappavaara and Vittangi gabbros all had U–Pb ages 1880 to 1860 Ma (Skiöld 1982). The Rb–Sr age of one of these rocks, a granite, as well as of another granite in the area is however much lower, 1725 ± 16 Ma (Skiöld 1982) and 1780 ± 49 Ma (Welin et al. 1971) respectively. This could be either a result of a resetting of the radiometric clock caused by serorogenic intrusions or a result of a very slow rate of uplift and cooling.

The inhomogeneity of the rock is well expressed in the silicate densities. Also the susceptibilities vary considerably and the Q-values, generally around 1.0 and less, are low in comparison with the previously discussed gabbros.

Contrary to the Svappavaara gabbro, the basic rock of the large (64 km^2) well rounded formation at *Vittangi* is quite homogeneous, although differentiates and possibly younger veins exist (Geijer 1918). The rock is quite fresh with few signs of alteration and secondary magnetite exists only in very small amounts. Exsolution lamellae of haematite in ilmenite are observed as well as ore symplectites and myrmekitic textures.

To the previous determinations we can also add a Rb–Sr age, 1707 ± 77 (Skiöld 1982), of a

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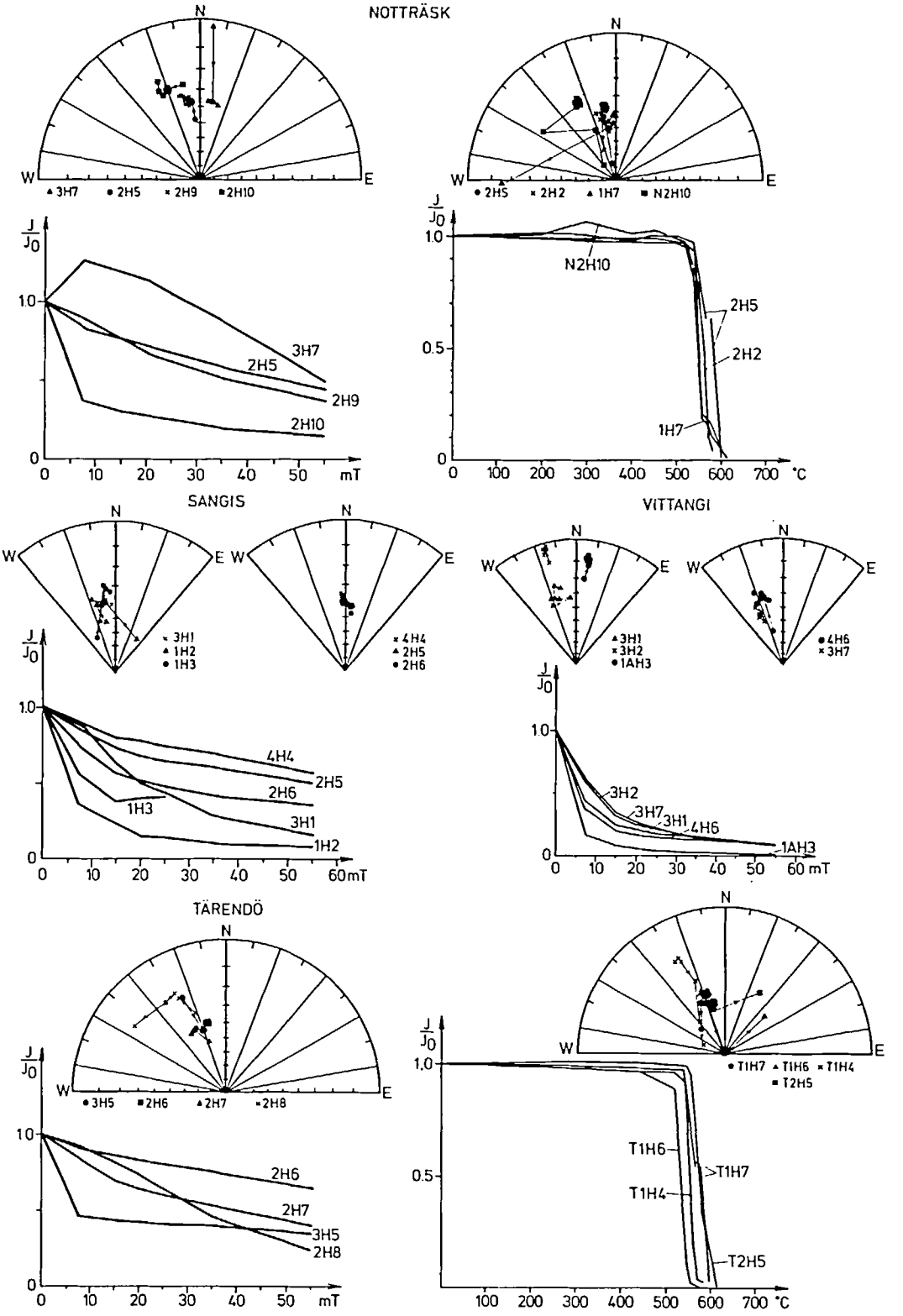


Fig. 3. Behaviour of directions of remanent magnetization with progresssive AF and thermal demagnetizations. The numbers refer to sample and site numbers.

Table 1. Palaeomagnetic results for the Notträsk gabbro (65°50'N, 21°50'E) at 25 mT AF.

Site	N	D	I	K	α_{95}	Palaeomagnetic pole		dp	dm
						°E	°N		
H1	2	318.4	-3.1	40		245.6	16.3		
H2	5	348.9	34.9	82	8.5	216.2	42.8	5.6	9.8
H3	2	348.1	35.1	217		217.2	42.9		
H4,5,6	7	349.1	38.2	31	11.0	216.2	45.1	7.7	13.0
H7	3	3.1	39.3	792	4.4	197.7	46.3	3.1	5.2
H9	5	342.9	35.7	83	8.4	223.9	42.5	5.7	9.8
H10	3	342.2	31.1	195	8.9	224.1	39.5	5.5	9.9
Mean of sites 2,3,5,9,10	5	346.2	35.0	460	3.6	219.7	42.6	2.4	4.1
Mean of samples	22	346.6	35.7	61	4.0	219.3	43.0	2.7	4.6

granodiorite south of the formation reinforcing the difference in U–Pb (intrusion age?) and Rb–Sr age. The susceptibility is homogeneously distributed throughout the rock and the Q-values are in general low, less than 1.0.

The *Dundret* formation, 42 km^2 in area, is an olivine–norite to norite gabbro with borderline facies of dioritic and syenitic composition (Geijer 1931). The rock is partly altered and secondary chlorite, amphibole and ore minerals are observed. Secondary magnetite occurs as very fine-grained lamellae in pyroxene crystals. The grain size of the primary magnetite, which occurs in relatively small amounts together with ilmenite, is definitely larger. Myrmekitic textures and a few ore symplectites are observed as well.

The rock has been radiometrically dated to 1890 and 1950 Ma (K–Ar, Magnusson 1970). Similar ages were obtained for volcanic rocks north of the formation, 1875 Ma (K–Ar, Welin et al. 1972) and 1909 ± 17 Ma (U–Pb, Skiöld & Cliff, in press). Granites further north of the gabbro were also dated and it is concluded that some of them, the Lina granites, have ages around 1530 ± 35 Ma (Rb–Sr) (Welin et al. 1971). According to the 'silicate densities' the most basic rock is in the central part of the formation where also most of the samples are taken. The susceptibilities and Q-values are widely dispersed. However the Q-values in general are low.

Palaeomagnetic results

Alternating field and thermal demagnetization

A soft component of remanence exists in most of the specimens and is generally erased after treatment in 7.5 to 15.0 mT AF. Changes in directions, from generally higher inclinations towards lower ones, indicate these soft components to be influenced by the earth's present magnetic field and thus probably of viscous origin.

The characteristic remanence of the *Notträsk* rock is relatively hard and stable (Fig. 3). The specimens from Site 7 situated in the margin of the highly magnetic ring structure show somewhat different magnetic properties. The characteristic remanence is not as hard as that found for most of the other specimens, but the direction is stable (Fig. 3). The mean direction of this site is also significantly different from the majority of the other site means. The poorly defined mean direction of Site 1, which is far from the mean of the rest of the sites, and the mean of Site 7 have been disregarded in estimating the formation mean. The specimens from the three sites 4, 5 and 6 are all sampled within a distance of about 700 meters and as their directions of remanence are similar they are treated together (Site 5'). Using the sites 2, 3, 5', 9 and 10, a formation mean with high precision is obtained at 25 mT AF (35° inclination and 346° declination, Table 1). In the Notträsk gabbro, as for all other formations in this investigation, the mean of site means is not significantly different from the mean of samples.

Thermal demagnetization was carried out on specimens from four sites (2, 5, 7, 10, Fig. 3). As the Curie temperature of magnetite is approached an increase in inclination is observed in some of the specimens. Taking the means of three of four specimens (2H5, 1H7, 2H10) demagnetized, an increase in inclination from 37.2° (525°C) to 45.6° (580°C) is noticed, while the declinations remain approximately the same.

The directions of the remanence vectors of the

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Table 2. Palaeomagnetic results for the Sangis gabbro (65°50'N, 23°20'E) at 35 mT AF.

Site	N	D	I	K	α_{95}	Palaeomagnetic pole		dp	dm
						°E	°N		
H1	3	356.2	37.3	326	6.8	208.4	44.9	4.7	8.0
H2	3	350.4	43.0	73	14.6	216.6	48.7	11.2	18.0
H3	1	–	–			–	–		
H4	4	2.9	38.0	436	4.4	199.5	45.5	3.1	5.2
H5	5	0.9	47.4	176	5.8	202.1	52.7	4.9	7.5
H6	3	356.1	40.3	623	4.9	208.6	47.0	3.6	6.0
H7	2	356.8	41.2	285		207.8	47.8		
H8	4	359.2	37.5	432	4.4	204.5	45.1	3.1	5.2
H9	3	352.5	40.5	610	5.0	213.6	47.0	3.7	6.0
H10	3	0.5	38.1	568	5.2	202.7	45.6	3.6	6.1
H11	2	2.8	34.0	45		199.8	42.8		
H12	3	4.0	36.1	220	8.3	198.2	44.1	5.6	9.7
Mean of sites	11	358.5	39.5	264	2.8	205.4	46.5	2.0	3.4
Mean of samples	35	358.7	39.9	141	2.1	205.2	46.9	1.5	2.5

Table 3. Palaeomagnetic results for the Tärendö gabbro (67°10'N, 23°30'E) at 55 mT AF.

Site	N	D	I	K	α_{95}	Palaeomagnetic pole		dp	dm
						°E	°N		
H1	5	338.8	44.8	647	3.0	231.0	47.2	2.4	3.8
H2	2	337.0	36.3	228		231.4	40.8		
H3	3	316.0	34.1	163	9.7	255.0	34.1	6.4	11.1
H4	4	339.4	20.3	32	16.6	226.5	31.7	9.1	17.4
H5	4	342.3	46.2	328	5.1	226.8	49.0	4.2	6.5
H6	3	343.0	40.1	580	5.1	224.7	44.5	3.7	6.2
H7	4	335.8	40.1	160	7.3	233.7	43.2	5.3	8.8
H8	2	305.3	13.3	216		261.7	19.3		
Mean of sites 1,2,5,6,7	5	339.3	41.5	303	4.4	229.6	44.9	3.3	5.4
Mean of samples	18	339.7	42.4	203	2.4	229.8	45.5	1.8	3.0

Sangis rock are generally very stable (Fig. 3). The remanence of all samples, with the exception of the rock of dioritic composition, is also quite hard. Two of the four dioritic sites (2 and 3) have considerable soft remanence, erased already at 7.5 mT AF. The precision of the site means is high and as the means are not significantly different from each other they are all used for the formation mean (Table 2).

No systematic change in the remanence vector is noticed as the demagnetizing temperatures approach 580°C. The lowest Curie temperatures are found among the specimens showing the most eastern declinations, with the specimens of Site 5 (K2H5) as exception. The existence of haematite at Site 11 is confirmed by the graph of demagnetization where 15 % of the residual remanence is still left at 580°C. However no stable direction is obtained.

The remanent magnetization of the *Tärendö* gabbro is in general very strong and stable (2.4–24.Am^{-1}) and apart from the sites 2 and 3 also quite hard. The two specimens from Site 8, a gneiss which is probably a granitized part of the gabbro (Pagdet, 1970), show a different graph of demagnetization and no stable endpoint is reached (Fig. 3). The precisions of the site means are generally high and not significantly different from each other. However, the somewhat more basic rocks at the sites 3 and 4 have mean directions different from some other sites. Excluding these two site means and site 8 a formation mean of high precision is obtained at 55 mT (41.5° melination, 339.3° declination, Table 3).

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Table 4. Directional changes of the remanent magnetization during thermal demagnetization of specimens from the Tärendö and Svappavaara gabbros.

	N/n	C°(± 10°)	D	I	K	α_{95}	E	N	dp	dm
Tärendö	4/4	545	339.1	39.4	33	16.3	229.5	43.3	12	20
		560	339.5	45.4	95	9.5	230.2	47.9	8	12
		570	336.0	52.4	112	8.7	237.2	53.1	8	12
		580	329.2	60.9	29	17.4	251.7	59.7	20	27
Svappavaara	4/4	500	329.3	48.1	79	10.4	242.4	47.3	9	14
		525	319.7	52.4	37	15.2	256.9	49.9	15	22
		540	317.7	55.1	53	12.7	259.7	50.1	13	18
		560	307.0	67.5	34	16.1	284.1	59.1	22	27

Table 5. Palaeomagnetic results for the Svappavaara gabbro (67°40'N, 21°10'E) at 20 mT AF.

Site	N	D	I	K	α_{95}	Palaeomagnetic pole		dp	dm
						°E	°N		
H1	5	350.3	62.9	98	7.8	218.5	66.0	9.5	12.2
H3	3	358.7	34.6	103	12.2	202.8	41.3	8.0	14.0
H4	5	330.9	46.4	80	8.6	239.8	46.3	7.1	11.1
H5	2	333.8	49.5	440		237.2	49.6		
H6	1	352.0	66.9			217.6	71.4		
H8	4	337.5	53.9	174	7.0	234.0	54.3	6.8	9.8
H9	3	332.7	54.1	285	7.3	240.5	53.0	7.2	10.3
H11	3	344.1	51.8	22	26.9	227.5	51.2	21.3	31.9
H4 ± H5	7	331.7	47.3	108	5.8	239.1	47.2	4.9	7.5
H1 ± H6	6	350.5	63.5	117	6.2	218.4	66.9	7.8	9.9
Mean of sites H4 ± H5, 8,9,11	4	336.4	51.9	293	5.4	234.6	52.2	5.0	7.3
Mean of samples	17	335.2	50.9	74	4.2	235.8	51.1	3.8	5.6

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Five specimens from different sites were thermally demagnetized. A change in directions from intermediate NNW to more steep NW is noticed in four of the five specimens when the temperature approaches 580°C (Fig. 3, Table 4). An interpretation of this is given in the next section. The fifth specimen, from site 3, is not stable during demagnetization.

In the *Svappavaara* formation, the intensity of remanence, like the susceptibility, varies considerably (0.03–10.8 Am^{-1}). Erasing the soft component, the characteristic remanence is quite hard and stable (Fig. 4) in all specimens except that only one specimen from the dioritic rock at Site 6 was strong and stable enough to be used. As the direction of remanence is within the cone of confidence of the dioritic rock at Site 1 and both sites are situated at the eastern margin, thereby possibly reflecting the same geological milieu, they are treated together. For similar reasons the specimens of Site 4 and 5 are treated as one site. At 25 mT AF the mean of the site means (4 + 5) 8, 9 and 11 shows a high precision (51.9° inclination, 336.4 declination, Table 5). The site means (1 + 6) and 3, significally different from three of the four other sites, show a more northern declination with a higher respectively lower inclination than the formation mean.

As in the case of the specimens from Tärendö, a general change in the directions of the remanence vectors during thermal demagnetization is observed. When the Curie temperature of magnetite is approached the directions are changed from intermediate NNW to more steep NW (Fig. 4). A mean of the four specimens, all taken from different sites, is calculated for temperatures increasing from 500°C to 560°C (Table 4).

A large proportion of the remanent magnetization of the *Vittangi* gabbro is soft and at least partly of viscous origin. This is illustrated by the directional changes and the graphs of demagnetization showing a decrease in intensity of sometimes as much as 85 % at 7.5 mT AF (Fig. 3). The characteristic remanence is relatively hard

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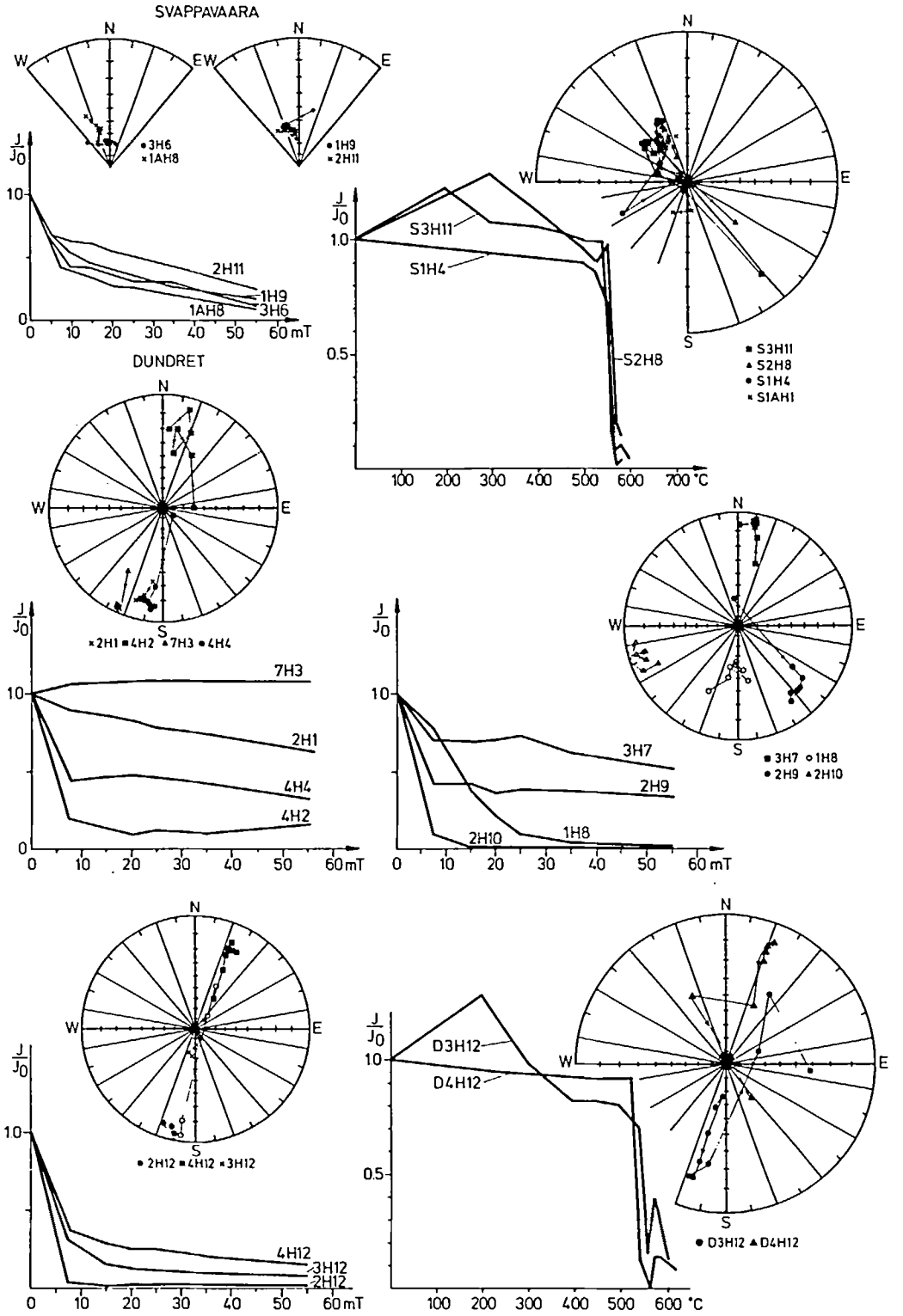


Fig. 4. Behaviour of directions of remanent magnetization with progressive AF and thermal demagnetization. Open symbols denote directions of the remanence vectors on the upper hemisphere.

Table 6. Palaeomagnetic results for the Vittangi gabbro (67°40'N, 21°40'E) at 25 mT AF.

Site	N	D	I	K	α_{95}	Palaeomagnetic pole		dp	dm
						°E	°N		
H1	5	343.7	32.3	19	18.0	221.8	38.7	11.4	20.3
H2	3	343.4	7.4	665	4.8	220.0	25.0	2.4	4.8
H3	5	342.8	4.1	29	14.5	220.5	23.3	7.3	14.5
H6	4	337.2	43.1	26	18.5	231.5	45.2	14.2	23.0
H7	3	336.6	46.3	2190	2.6	233.1	47.5	2.2	3.4
H8	4	344.4	30.6	54	12.6	220.8	37.8	7.8	14.1
H9	3	325.7	36.3	63	15.7	243.7	37.9	10.6	18.3
H10	3	340.7	36.4	16	31.7	226.0	41.0	21.5	37.0
H12	3	339.9	41.2	126	11.6	227.7	44.3	8.2	13.4
H13	3	347.3	45.0	26	24.6	218.8	48.2	19.7	31.1
Mean of sites 1,6,7,8,9,10,12,13	8	339.5	39.1	107	5.4	227.9	42.6	3.8	6.4
Mean of samples	28	339.9	38.5	32	4.9	227.3	42.3	3.5	5.7
Mean of sites (2,3)	2	343.1	5.8	117		220.3	24.2		
Mean of samples	8	343.0	5.4	48	8.0	220.3	24.0	4.0	8.1

and stable. Two of the site means (site 2 and 3) are significantly different from the other mean directions. However the inclinations increase, moving the means closer to the remaining site means, after treatment in higher AF. These two means are not included in the formation mean. The eight remaining site means form a mean which reaches its highest precision at 25 mT AF (39.1° inclination, 339.5° declination, Table 6). No systematic changes in directions were noticed during the thermal demagnetization. The presence of haematite is indicated in some specimens by the fraction of remanence not erased above the Curie temperature of magnetite.

The characteristic remanence of the *Dundret* gabbro is in general hard but not so stable as that of the previously described gabbros. The directional differences are quite large between the sites and sometimes also within the sites. If the poorly defined means of the sites 2, 9 and 10 are disregarded, the remanence directions are however confined approximately to a great circle passing through 10° NNE. Four of these six remaining sites (1, 3, 4 and 12) have remanences of very low inclination and a polarity opposite to the polarities of the previously considered formations in this study. Site 8, showing a different graph of demagnetization (Fig. 4), has a well defined mean of opposite polarity along the great circle but with a higher inclination. Only one site (7) shows a normal polarity (Fig. 4). Specimens of opposite polarity are found also within the same sites (7 and 12) and during progressive demagnetization the directions of the remanence vectors of some specimens move along the great circle (Fig. 4). A similar behaviour is observed during the thermal demagnetization of specimens from Site 12 (Fig. 4). A large part of the residual remanence of specimen D3H12 is of low Curie temperature origin. On reaching 400°C the directions become fairly stable just to change again moving back along the great circle when approaching the Curie temperature. The distribution of the directions of the remanence vectors along the great circle can be explained by the presence of the two generations of magnetite. The differences in grain size of the two phases result in different hardness of remanence. At some sites and in some specimens one of the phases is dominating the other, while in others vector addition of the components gives directional variations along the great circle.

The precision of the individual site means varies but is in general not very high. The means of the sites 7, 8 and 10 are significantly different from each other and from all other means and therefore treated separately. At 35 mT a relatively well defined formation mean of reversed polarity (2.5° inclination, 192.5° declination, Table 7) is obtained, using the means of sites 1, 3, 4 and 12.

Palaeomagnetic pole positions and age of magnetization

It is concluded from the thermal studies that the blocking temperature range of the individual sites is wide enough to average secular vari-

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Table 7. Palaeomagnetic results for the Dundret gabbro (67°05'N, 20°35'E).

Site	N	D	I	K	α_{95}	Palaeomagnetic pole		dp	dm	
						°E	°N			
H1	4	194.8	4.8	94	9.5	4.9	−19.7	4.8	9.5	35 mT
H2	3	57.2	61.9	5	65.7	119.8	51.5	78.7	101.7	20 mT
H3	5	196.7	1.2	167	5.9	2.6	−21.3	3.0	5.9	35 mT
H4	4	182.7	1.0	45	13.9	17.7	−22.3	6.9	13.9	35 mT
H7	5	14.7	12.6	24	16.0	183.9	28.4	8.3	16.3	35 mT
H8	5	175.0	−48.0	88	8.2	27.7	−51.8	7.0	10.7	25 mT
H8'	3	201.1	−34.1	14	34.9	354.3	−39.7	22.8	40.0	55 mT
H9	3	164.3	14.4	6	56.5	36.7	−14.7	29.6	57.9	25 mT
H10	3	283.5	3.0	13	36.2	278.7	6.6	18.1	36.3	25 mT
H12	2	195.7	3.0	111		3.8	−20.5			35 mT
Mean of sites 1,3,4,12	4	192.5	2.5	143	7.7	7.2	−21.1	3.9	7.7	35 mT
Mean of samples	15	192.4	2.4	65	4.8	7.3	−21.2	2.4	4.8	35 mT

ations. It seems therefore that the site means in this study can be regarded as palaeomagnetic poles.

The poles of the sites in the *Nottråsk* gabbro, apart from Site 7 and the poorly defined Site 1, are grouped quite close together giving a formation pole position at 42.6°N and 219.7°E (Fig. 5). The pole of Site 7 is at about the same latitude but 20° further west. The rock at this site is metamorphosed and as secondary magnetite is present it is at least partly remagnetized. A high-temperature component, in strength 20 % of the total intensity and in direction similar to the other sites, is also shown by the thermal study. The rock at site 1 is affected by complex metamorphic processes (Elming 1982) and its mean is not considered to be a meaningful pole position.

An interpretation of the gravimetric and magnetic anomalies in the area indicates the gabbro to be somewhat tilted from its original position (Widenfalk et al., in press). If the tilting, approximately 5°, has occurred after the rock has been magnetized the mean pole of the formation should be moved from 42.6°N and 219.7°E to 45.9°N and 217.2°E. This difference is almost within the error of pole determination. As this formation probably belongs to the Haparanda suite a maximum age of magnetization is 1840 Ma (Welin et al. 1970). As the rock is at least partly remagnetized, possibly by the intrusion of the serorogene–postorogene Edefors granite, the direction of the polar wandering is from east to west. Using the apparent polar wandering speed of 0.35°/Ma for the Baltic shield (Pesonen & Neuvonen 1981) this metamorphic part is found to have been remagnetized 55 to 75 Ma later than the main part of the formation.

The site mean directions in the *Sangis* gabbro, varying from intermediate NNW to more shallow inclinations at NNE, correspond to pole positions scattered from 217°E to 198°E longitude and 43°N to 53°N latitude (Fig. 5).

Also this massif belongs to the Haparanda suite, which means that the maximum age of magnetization is 1840 Ma. As the existence of haematite does not reveal any separate magnetization component and secondary magnetite is definitely subordinate to the primary one, the highest Curie temperatures found among the most eastern poles should represent the oldest magnetizations. This supports the idea of migration of the palaeofield from east to west.

In the *Tärendö* gabbro the site pole positions are rather well concentrated and the sites 1, 2, 5, 6 and 7 form a mean with high precision at 230°E and 45°N (Fig. 5). Although the massif is deformed by folding and faulting the site means are not significantly different from each other. The low inclinations of the sites 3 and 4 are increased during the thermal demagnetization. The general change in directions from intermediate NNW to steep NW when the temperatures approach 580°C indicate a backward continuation of the polar wandering-path towards ENE (Fig. 9, Table 4). The recording of the apparent polar-wandering within such hard and stable remanence specimens (single domain?) is possible since it has been shown that the blocking temperature ranges are broadened and lowered by slow cooling (Pullaiah et al. 1975; Dodson & McClelland-Brown 1980). The very well developed ore symplectites support the hypothesis of slow cooling at least in a late stage of the evolution of the rock. A minimum age of the magnetization is

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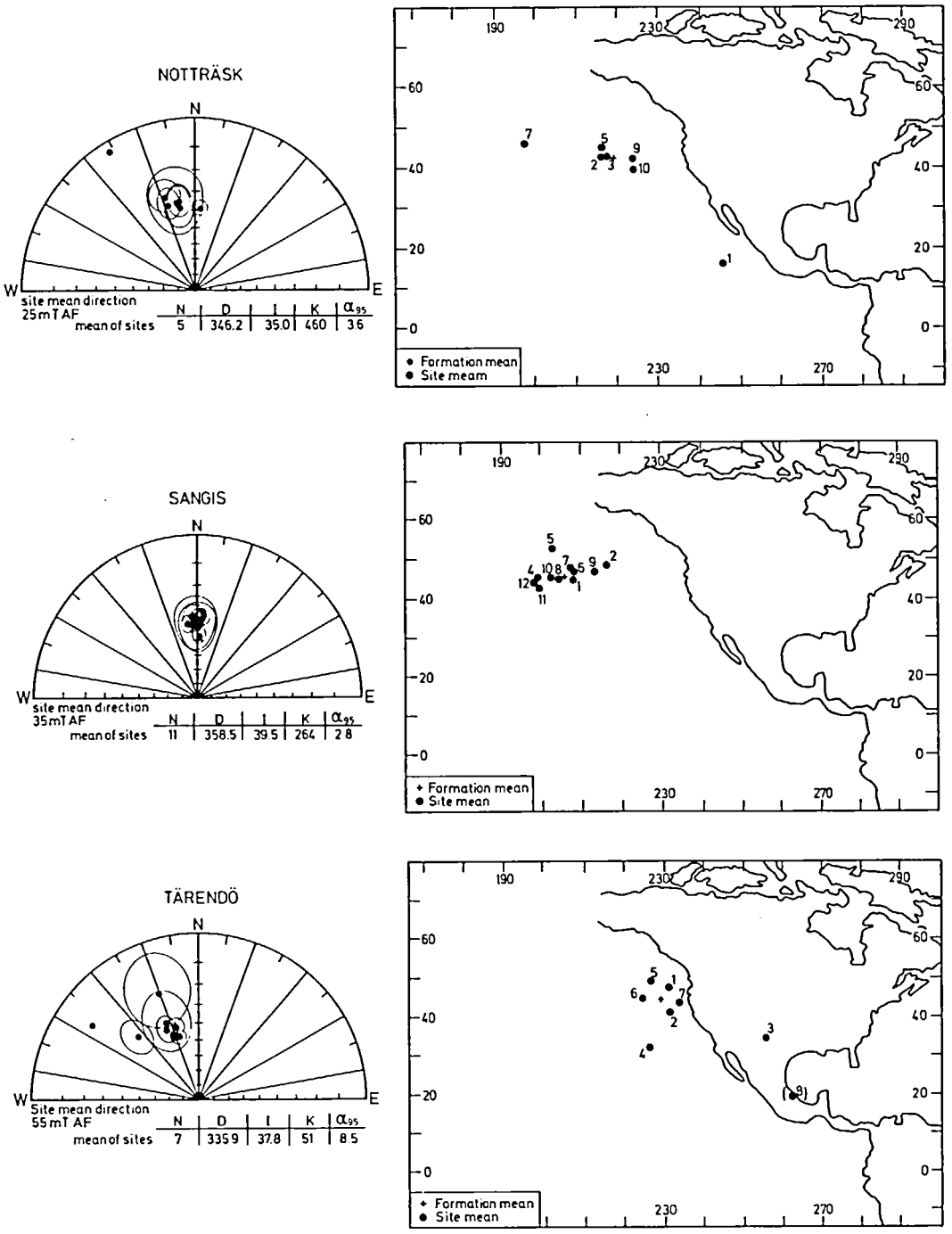


Fig. 5. Site mean directions with cone of confidence as well as apparent site mean pole positions of the Notträsk, Sangis and Tärendö formations. The mean apparent palaeomagnetic pole of a formation is marked by a cross.

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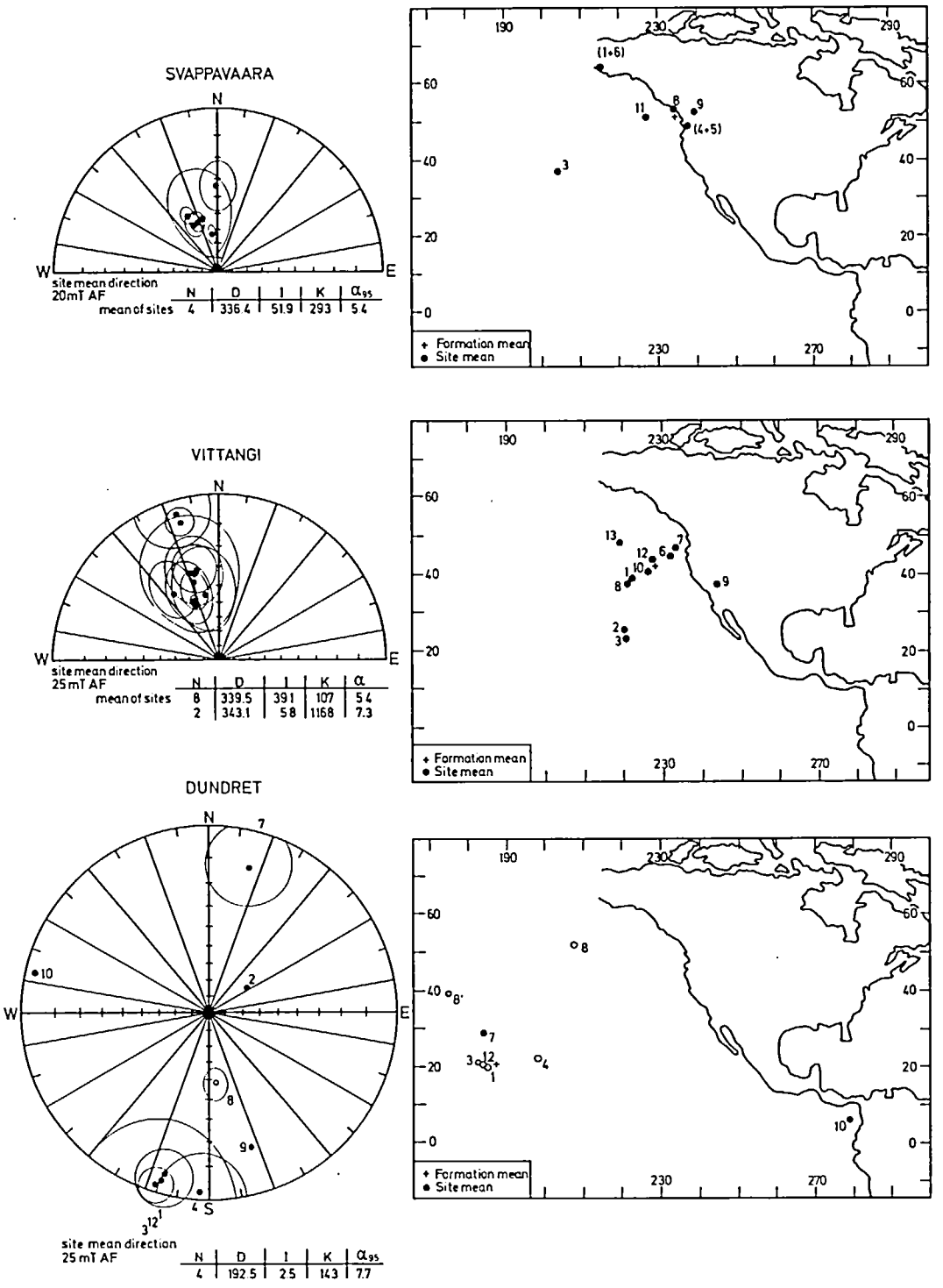


Fig. 6. Site mean directions with cone of confidence as well as apparent site mean pole positions of the Svappavaara, Vittangi and Dundret formations. The mean apparent palaeomagnetic pole of a formation is marked by a cross.

related to the age of the surrounding granites ~1780 Ma.

Apart from the poles of the anomalous sites 3 and (1 + 6), the apparent palaeomagnetic poles of the *Svappavaara* gabbro are rather well concentrated forming mean with high precision at 235°E and 52°N (Fig. 6, Table 5).

The rather unstable remanence of Site 3 gives a pole position 30 degrees west of the formation mean (Fig. 6), a position similar to the poles of the basic rocks at Sangis and the single 'metamorphic' pole of Notträsk. As poles of younger rocks are found further west the rock at site 3 seems to have been magnetized at a later orogenic stage.

The directional changes during the stepwise thermal demagnetization indicate a backward extension of the apparent polar wandering path quite similar to that found for the Tärendö gabbro. The difference in radiometric ages of the surrounding rocks, U–Pb = 1880–1860 Ma and Rb–Sr 1780, 1725 Ma is either a proof of a very slow rate of uplift and cooling or an influence of a serorogene metamorphic event. As for the Tärendö gabbro a slow cooling of the Svappavaara gabbro is indicated by myrmekite textures. The similarity in pole position and thermal behaviour of these two gabbros indicates them to be magnetized at the same time and under similar circumstances. The age of magnetization is therefore to be found in the time range 1880 ~ 1725 Ma. The pole position of the well-defined mean of Site 1 (and 6) situated at a latitude, not obtained from any other formation cannot be explained at present.

Also in the *Vittangi* gabbro an east–west distribution of the apparent site mean pole position is obvious (Fig. 6, Table 6).

Contrary to the Svappavaara and Tärendö gabbros the thermal demagnetization of Vittangi samples does not show any high Curie temperature components that can extend the path eastward. The well rounded shape of this massif, in contrast to the shapes of the Svappavaara and Tärendö massifs, might indicate a later intrusion of this rock and therefore no record of older palaeofield similar to that of the other two rocks is obtained. The same mineralogical signs of slow cooling are also found in this rock and the age of magnetization is probably similar but somewhat younger than for Svappavaara and Tärendö.

The reversely polarized site means (1, 3, 4 and 12) and the Site 7 of normal polarity at *Dundret* are well grouped together but found at a lower latitude and a more westerly longitude than the previously discussed site means (Fig. 6). The

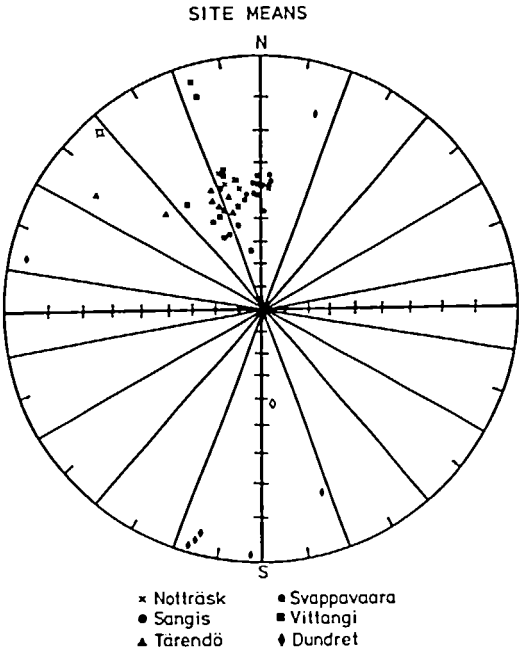


Fig. 7. Site mean directions of all sites in this study.

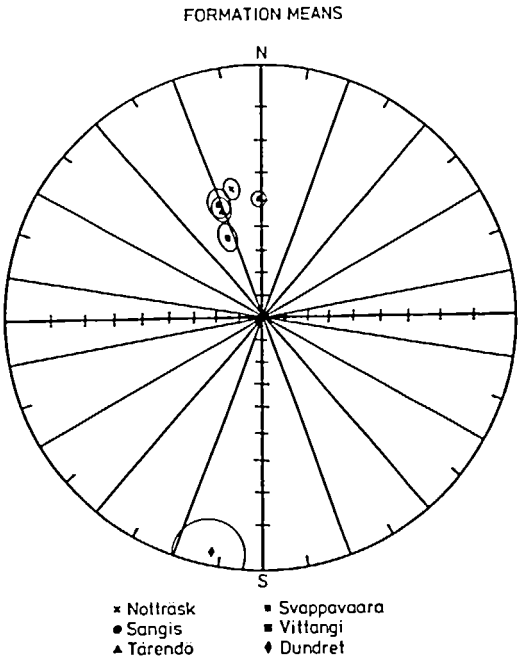


Fig. 8. Mean directions and cone of confidence of site means in the different rock formations.

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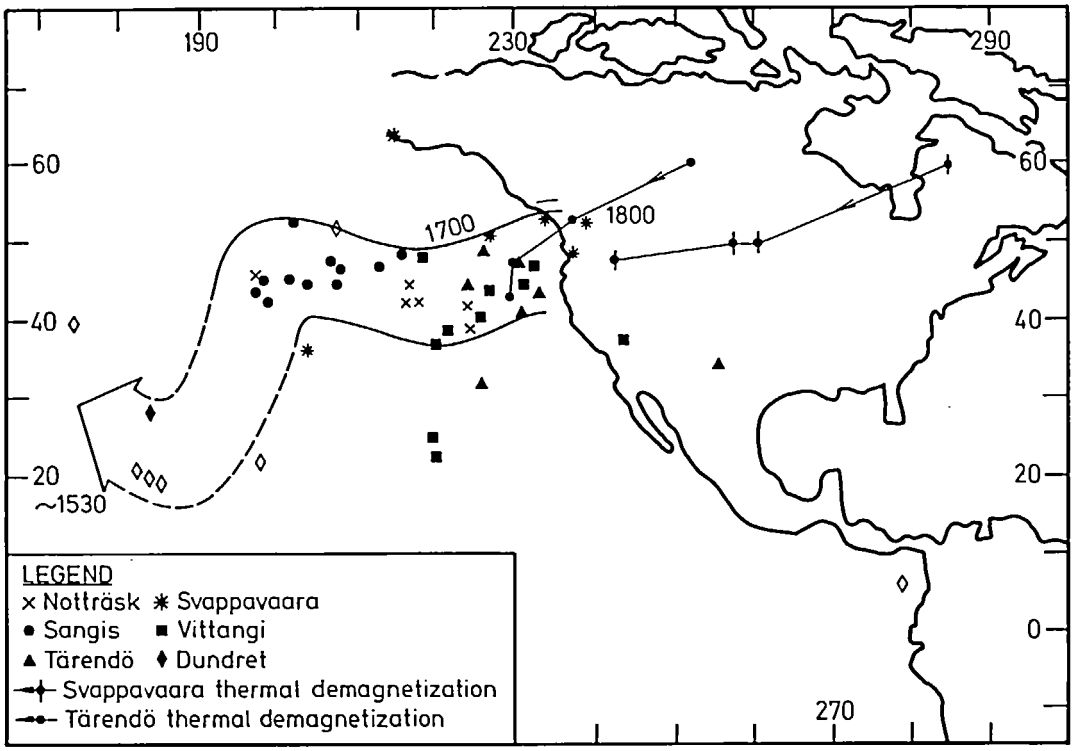


Fig. 9. Apparent palaeomagnetic pole positions of all site means in this study. The solid swathe indicates an apparent polar wandering path from about 1800 Ma to 1600 Ma as a 12° wide band. The dashed line shows a probable continuation of the polar wandering path to about 1530 Ma. A possible backward continuation is indicated from the thermal demagnetizations.

pole of Site 8 is situated close to the site means of the Sangis gabbro, while the pole position of Site 10 is far away (at 278.7°E 6.6°N) from the other poles in this study. Three of the specimens from Site 8 give after further demagnetization high coercivity components with a pole (40°N, 174°E) somewhat closer to the formations' mean.

The very poorly defined means of the sites 2 and 9 are not regarded as meaningful poles. Although radiometric ages of this gabbro (K–Ar; 1890, 1950 Ma) and the immediately surrounding rock (K–Ar; 1875, U–Pb; 1909) are the same as those of the other gabbros, the pole positions are very different. The alteration of the rock and the presence of a large amount of secondary magnetite indicate that a remagnetization has taken place. The only younger rocks reported in the area are the younger Lina granites intruded about 1530 Ma ago. A remagnetization associated with the intrusion of these granites is at present the only possible explanation of these pole positions.

Discussion of results

Looking at the distribution of the site pole positions of the individual formations (Fig. 9) it seems clear that some of the poles are a result of remagnetization. The metamorphosed rock at Notträsk is such an example giving a pole position 20 degrees west of the other poles. This pole is similar to the poles of the Sangis gabbro, even if the formation means of the two are significantly different (Fig. 8). Consequently, as both gabbros belong to the Haparanda suite, the Sangis rocks are either reheated and remagnetized or represent a deeper erosion level, cooled at a later stage than the Notträsk gabbro.

The site means of the Notträsk and Vittangi rocks overlap to some extent, the Vittangi site means extending the pole distribution further east. Although surrounded by rocks of the same age (1880–1860 Ma, U–Pb, 1725–1707, Rb–Sr) the mean pole positions of the Svappavaara and Vittangi gabbros are significantly different. The

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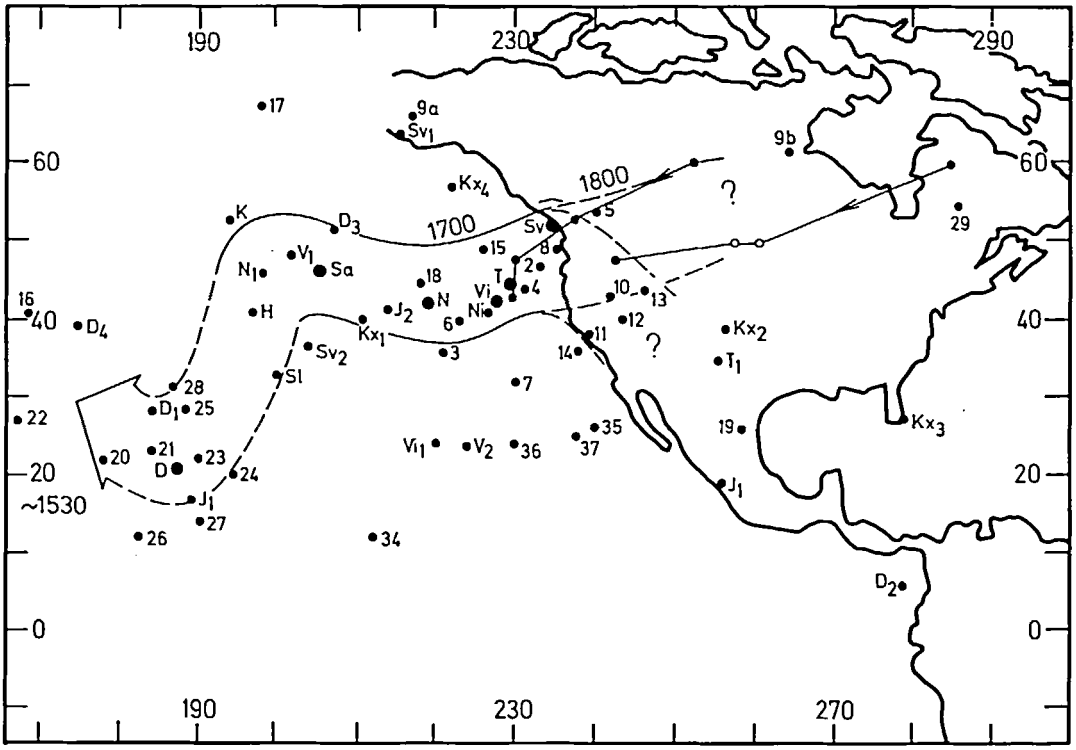


Fig. 10. Apparent palaeomagnetic pole positions from the Fennoscandian shield. Two possible directions of a backward continuation of the polar wandering path are indicated.

difference in density and magnetic properties as well as the shape of the formations may indicate genetical differences. The Svappavaara gabbro and the Tärendö gabbro are both deformed and irregularly shaped in contrast to the Vittangi and Notträsk rocks. It seems likely that these rocks are older than the surrounding rock while the Vittangi and Notträsk are cogenetic or younger. This explains why a possible extension of the polar wandering path is obtained only in these irregularly shaped massifs.

Very different from these continuously distributed poles are the site means of Dundret. The reversed and normally polarized site means of this formation are scattered and quite different from the other formation means. Radiometric datings of the gabbro and the surrounding rocks indicate the Dundret massif to be of Haparanda age. Still, the formation mean pole position is very different from the previous rocks and is more to be compared to younger rocks from central Sweden as for example the Loftahammar gabbro (Porter 1976) the 'Upper' Dala volcanics (Mulder 1971) and the Hällefors Giant Dyke (Piper 1980) all of which are younger than 1660 Ma.

Comparison with other palaeomagnetic data and conclusions

Comparing the palaeomagnetic poles of this study with previously estimated ones in the area (Cornwell 1968; Piper 1980) they agree rather well with each other (Fig. 10, Table 8). However, in the study by Piper it was not possible to combine the site means of the Dundret rocks to give any formation mean and only one of the site means agrees with the formation mean of the massif in this study. Also in the Sangis rock there are some discrepancies.

The poles of the basic rocks of Sangis (Sa), Korsträsk (K), Harads (H), Vuollerim (V),

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Table 8. A compilation of Lower Proterozoic palaeomagnetic data from the Fennoscandian shield.

Pole	Rock unit	Palaeomagnetic pole °E	°N	dp/dm	Age (Ma)	Reference	
pre-Svecokarelian rocks (Finland)							
1	Nilsiä diorites	313	64	7/8	2684 (U–Pb)	Neuvonen et al. 1981	*in* Pesonen and Neuvonen 1981
2	Jatulian dikes and greenstones	233	47	11/18	2150 (U–Pb)	Neuvonen 1975	
3	Syöte gabbro	221	36	2/4	2440 (U–Pb)	Neuvonen 1975	
pre-Svecokarelian rocks (Soviet Union)							
4	Pukhta-Pedaselsk group	231	44	3/5	1950–1850 (K–Ar, U–Pb-Th)		
5	Shoksha group	240	54	3/5	1950–1850 (K–Ar, U–Pb-Th)		
6	Kola gabbro-norites	223	40	9/16	1880–1820	McElhinny & Cowley 1977	*in* Pesonen and Neuvonen 1981
7	Onega gabbro-diabases	230	32	1/2	1800–1600		
8	Central Karelia dikes	235	49	5/7	1870–1610		
9 a	Northern Karelia sandstones	217	65	8/10	1870–1610		
9 b	Central Karelia sandstones	264	62	5/7			
Svecokarelian rocks (Finland)							
10	Ylivieska gabbro	242	43	5/8	1885 (U–Pb)	Pesonen & Stigzelius 1972	
11	Pohjanmaa intrusives	239	38	8/15	~ 1900	Pesonen & Stigzelius 1976	
12	South Finland intrusives	243	40	10/18	~ 1900	Neuvonen 1974	*in* Pesonen and Neuvonen 1981
13	Hyvinkää gabbro	246	44	5/9	1875 (U–Pb)	Puranen 1980, Härme 1978	
14	Pielavesi gabbro	238	36	2/4	~ 1900	Pesonen, unpublished Helovuori 1979	
15	Rantasalmi lamprophyres	226	49	2/3	1837 (U–Pb)	Neuvonen et al. 1981	
16	Åva intrusives	169	41	9/17	1815–1803 (U–Pb)	Neuvonen 1970, Pesonen 1981	
17	Luontarivesi granodiorite	198	67	10/17	1790–1765 (U–Pb)	Neuvonen 1978	
18	Keuruu dikes	218	45		~ 1880	Pesonen et al., unpublished	
19	Keuruu dikes	258	26		≤ 1900	Pesonen et al., unpublished	
Svecokarelian basic rocks, (northern Sweden)							
Kx_1	Kallax gabbro	211	40	2/3			
Kx_2	Kallax gabbro	257	39	4/6			
Kx_3	Kallax gabbro	279	27	6/10			
Kx_4	Kallax gabbro	222	57	3/5			
K	Korsträsk gabbro	194	53	6/9			
Ni	Niemisel gabbro	227	41	6/10	~ 1840		
SL	Stora Lulevattnet gabbro	200	23	4/8	~ 1530	Piper 1980	
J_1	Jokkmokk basic rocks	256	19	6/12			
J_2	Jokkmokk basic rocks	189	17	7/13			
J_3	Jokkmokk basic rocks	214	41	3/5			
V_1	Vuollerim gabbro	202	48	15/25			
V_2	Vuollerim gabbro	224	24	5/10			
H	Harads amphibolites	197	41	17/30			
N	Notträsk gabbro	219.7	42.6	2.4/4.1	~ 1840		
N_1	Notträsk gabbro	197.7	46.3	3.1/5.2	~ 1840		
Sa	Sangis gabbro	205.4	46.5	2.0/3.4	1840 ± 390 (Rb–Sr)	This study	
T	Tärendö gabbro	229.6	44.9	3.3/5.4	> 1780		
T_1	Tärendö gabbro	255	34.1	6.4/11.1	> 1780		

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Pole	Rock unit	Palaeomagnetic pole °E	°N	dp/dm	Age (Ma)	Reference
Sv	Svappavaara gabbro	234.6	52.2	5.0/7.3	1880–1860 (U–Pb) 1725 ± 16 (Rb–Sr)	This study
Sv_1	Svappavaara gabbro	218.4	66.9	7.8/9.9	1880–1860 (U–Pb) 1725 ± 16 (Rb–Sr)	
Sv_2	Svappavaara gabbro	202.8	41.3	8.0/14.0	1880–1860 (U–Pb) 1725 ± 16 (Rb–Sr)	
Vi	Vittangi gabbro	227.9	42.6	3.8/6.4	1880–1860 (U–Pb) 1707 ± 77 (Rb–Sr)	
Vi_1	Vittangi gabbro	220.3	24.2	3.7/7.3	1880–1860 (U–Pb) 1707 ± 77 (Rb–Sr)	
D	Dundret gabbro	187.2	21.1	3.9/7.7	1875, 1890, 1950 (K–Ar), ~ 1530	
D_1	Dundret gabbro	183.9	28.4	8.3/16.3	1875, 1890, 1950 (K–Ar), ~ 1530	
D_2	Dundret gabbro	278.7	6.6	18.1/36.3	1875, 1890, 1950 (K–Ar), ~ 1530	
D_3	Dundret gabbro	207.7	51.8	7.0/10.7	1875, 1890, 1950 (K–Ar), ~ 1530	
D_4	Dundret gabbro	174.3	39.7	22.8/40.0	1875, 1890, 1950 (K–Ar), ~ 1530	
post-Svecokarelian rocks, Sweden						
20	Lofthammar gabbro	178	22	1/2	1655 ± 35 (Rb–Sr)	Porter 1976
21	N. Dalarna porphyries	184	23	–	1634 ± 38 (Rb–Sr)	Mulder 1971
22	Hällefors giant dyke A	167	27	5/9	1518	Piper 1980
post-Svecokarelian rocks, Finland						
23	Kuisaari dolerite	190	22	7/14	1650 (U–Pb)	Neuvonen 1977
24	Åland rapakivi	194	20	15/18	1659–1589 (U–Pb)	Johnson 1979
25	Föglö-Sottunga dikes	188	28	6/13	1523–1518 (Rb–Sr, U–Pb)	Pesonen & Suominen, unpublished
26	Kumlinge-Brändö dikes	182	12	5/9	1602–1556 (Rb–Sr)	Pesonen & Suominen, unpublished
27	Kumlinge dikes	190	14	8/14	1602–1556 (Rb–Sr)	Neuvonen 1970
28	Föglö dikes	187	31	6/12	1523–1518 (Rb–Sr, U–Pb)	Neuvonen et al. 1969
						in Pesonen and Neuvonen 1981
Ukrainian Shield						
29	Bug-Bodolsky complex	286	54		2300–2000 (K–Ar, Rb–Sr, U–Pb)	
30	Ingulets complex	325	64		2000–1700 (K–Ar, Rb–Sr, U–Pb)	
31	Ingulets complex	316	57		2000–1700 (K–Ar, Rb–Sr, U–Pb)	
32	Bokovyansk intrusion	301	52		1700–1500 (K–Ar, Rb–Sr, U–Pb)	
33	Takova granite	326	51		1600–1400	Kruglyakova (1961)
34	Turchinki gabbro	212	12		1400–1200	Spall (1973)
35	Uman granite	240	26		1750 (K–Ar, Rb–Sr, U–Pb)	Semenko (1968)
36	Korosken complex	232	24		1750 (K–Ar, Rb–Sr, U–Pb)	
37	Korosken complex	238	27		1750 (K–Ar, Rb–Sr, U–Pb)	

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Jokkmokk (J) and partly Notträsk (N), all surrounded by probably serorogene and primorogene rocks, are situated predominantly in the western part of the sequence of the polar wandering path (Fig. 13). The main pole of the Stora Lulevattnet (SL), is found at a somewhat lower latitude. This rock is supposed to be cogenetic with the Sjaunja syenite (Ödman 1957) which has a radiometric age of 1530 ± 25 Ma (Rb–Sr, Gulson 1972). The age of magnetization of this rock is therefore related to the postorogene stage and similar to what is thought to be the age of the mean pole of Dundret. The above mentioned rocks are all probably magnetized in a late serorogene to postorogene stage.

Comparing the pole positions from the northern part of Sweden with poles from other parts of the Baltic shield the poles of Dundret, Jokkmokk and Stora Lulevattnet are correlatable to the poles of partly older rocks from the central parts of Sweden and the post-Svecokarelian rocks from Finland (Fig. 10, Table 8). The primorogene to serorogene age of magnetization of the Svappavaara, Vittangi, Tärendö and Notträsk rocks is well supported by the pole positions of the Svecokarelian and pre-Svecokarelian rocks from the Soviet Union and Finland (Table 8). This consistency of palaeomagnetic data from the different tectonic regions indicate that the Baltic Shield has acted as a coherent unit during this time sequence.

The site mean directions running from relatively steep north-north-west to more shallow north-north-east inclinations (and south-south-west) (Fig. 7) give a palaeomagnetic pole distribution running from east to west and south-west (Fig. 9). The east–west pole distribution is observed also within some of the formations in this study (Notträsk, Sangis, Vittangi). The remanences in these formations are very stable. As no systematic movement of the remanence vector is observed during thermal demagnetizations incomplete cleaning is probably not the cause of this distribution. At two other formations, Svappavaara and Tärendö, high blocking temperature components of increasing inclinations have been obtained. Haematite is not observed in these rocks and therefore the components are referred to as the oldest magnetizations. As the uplift rates of pre-1000 Ma mobile zones were slow (Watson 1976) and very few signs of alteration exist in these two formations a slow rate of cooling, broadening and lowering of the blocking temperature range, can explain the directional changes.

The difference in radiometric age determinations generally giving 100 to 150 Ma lower Rb–Sr ages than U–Pb ages might also be explained by a slow cooling rate. As the U–Pb and Rb–Sr systems have different closing temperatures they should show quite different ages in a very slowly cooled terrain.

In any case the trend in directional changes, during thermal demagnetization, as well as in site distribution, is probably a record of the polar movement during the interval of uplift and cooling of the individual formations and of the province as a whole. Using an apparent polar wandering speed of 0.35°/Ma (Pesonen & Neuvonen 1981) the apparent polar wandering path shown in this study covers a time sequence of about 200 Ma. The indications of the eastward extension of the polar wandering path are however somewhat contradictory. Formation mean and some single site means of Svecokarelian rocks may indicate a south eastward continuation of the path, while the thermal demagnetization of the Svappavaara and Tärendö gabbros shows an east-northeastward continuation. The latest proposed extension is in accordance with that shown by Pesonen & Neuvonen (1981), who based their model on pre-Svecokarelian data from the Soviet Union and data from the Ukrainian shield (Fig. 10). Even if an east-northeast-ward extension seems likely, available pre-Svecokarelian data are yet too scanty to make any definite conclusion possible. It should be noticed that the Svecokarelian section of the apparent polar wandering path shown in this study is quite different from that suggested by Pesonen & Neuvonen (1981).

The change in direction of the remanence from that of the NRM during AF demagnetization, generally a decrease in inclination, is often rather small. Although there is a large difference in the precision of the mean the NRM directions can give some idea of the direction of the characteristic magnetization.

Looking at the geographical distribution of the sampled formations and the palaeomagnetic pole distribution it appears that the western poles are obtained predominantly from the samples in the western and southern parts of the entire northern province studied (Elming 1982). The young ages of magnetization of the Dundret and Stora Lulevattnet rocks and also of the Akkavaara gabbro indicated by the NRM (Henkel 1981) are explained by the existence of young rocks in this area, while the late magnetization of the primorogenic–serorogenic rocks in the south-western and southern parts indicates a later uplift of this province – or a reheating of the rock at a serorogene or postorogene stage.

Acknowledgements. – The author wishes to thank Dattatray Parasnis, for critically reading the manuscript, Lennart Widenfalk, for help in the microscope studies, Herbert Henkel and Dan Nisca (Geological Survey of Sweden) for making existing petrophysical data available. Thank are also due to Roger Lindfors, for help during fieldwork and in the preparation of samples, to Eva Walldén, for help in measurements, to Eva Aukea, for efficient typing of the text and to Milan Vnuk, for preparing the drawings. This work was financially supported by the Swedish National Science Research Council, (NFR).

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